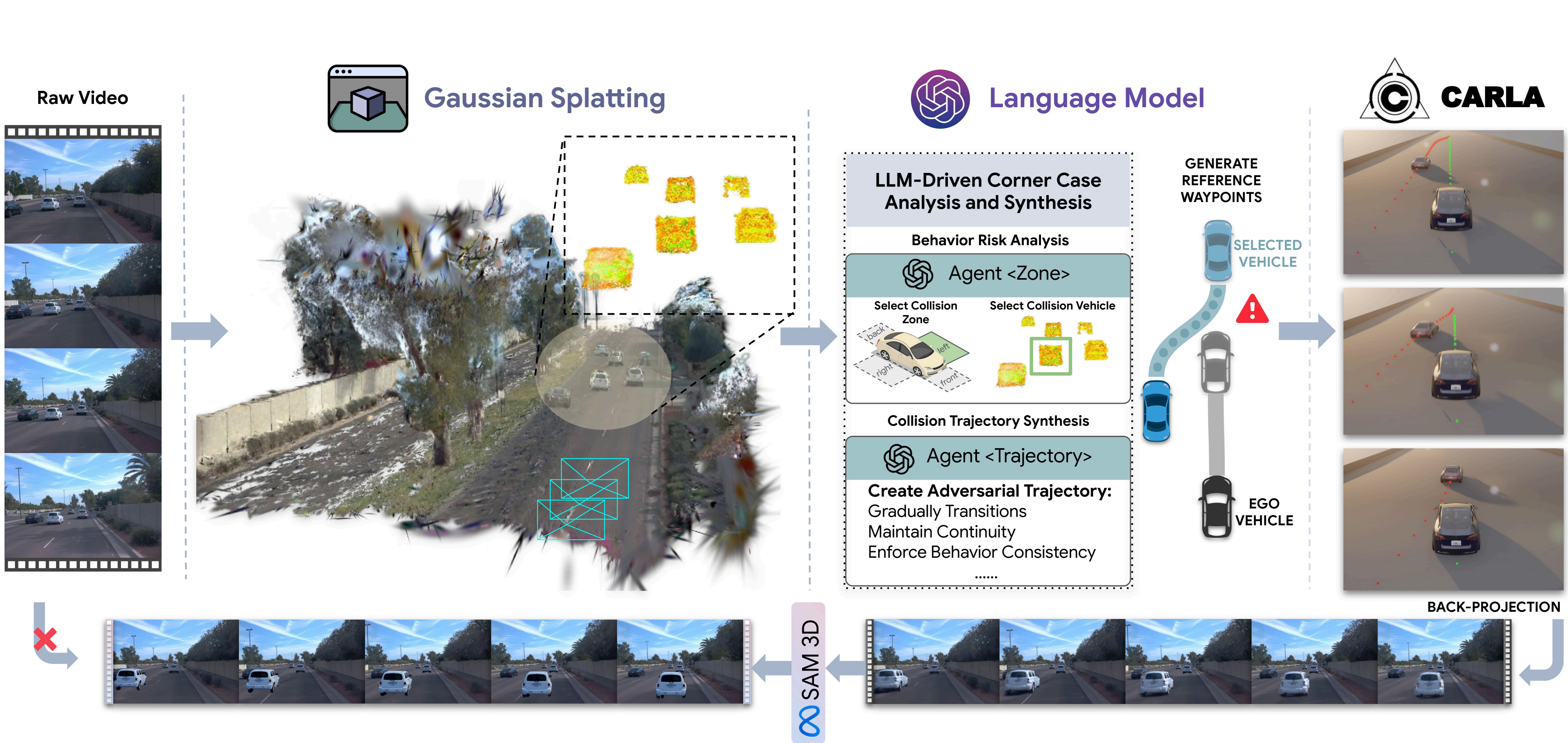


Decoupling Representation, Reasoning, and Physics Simulation for Autonomous Driving Corner-Case Synthesis

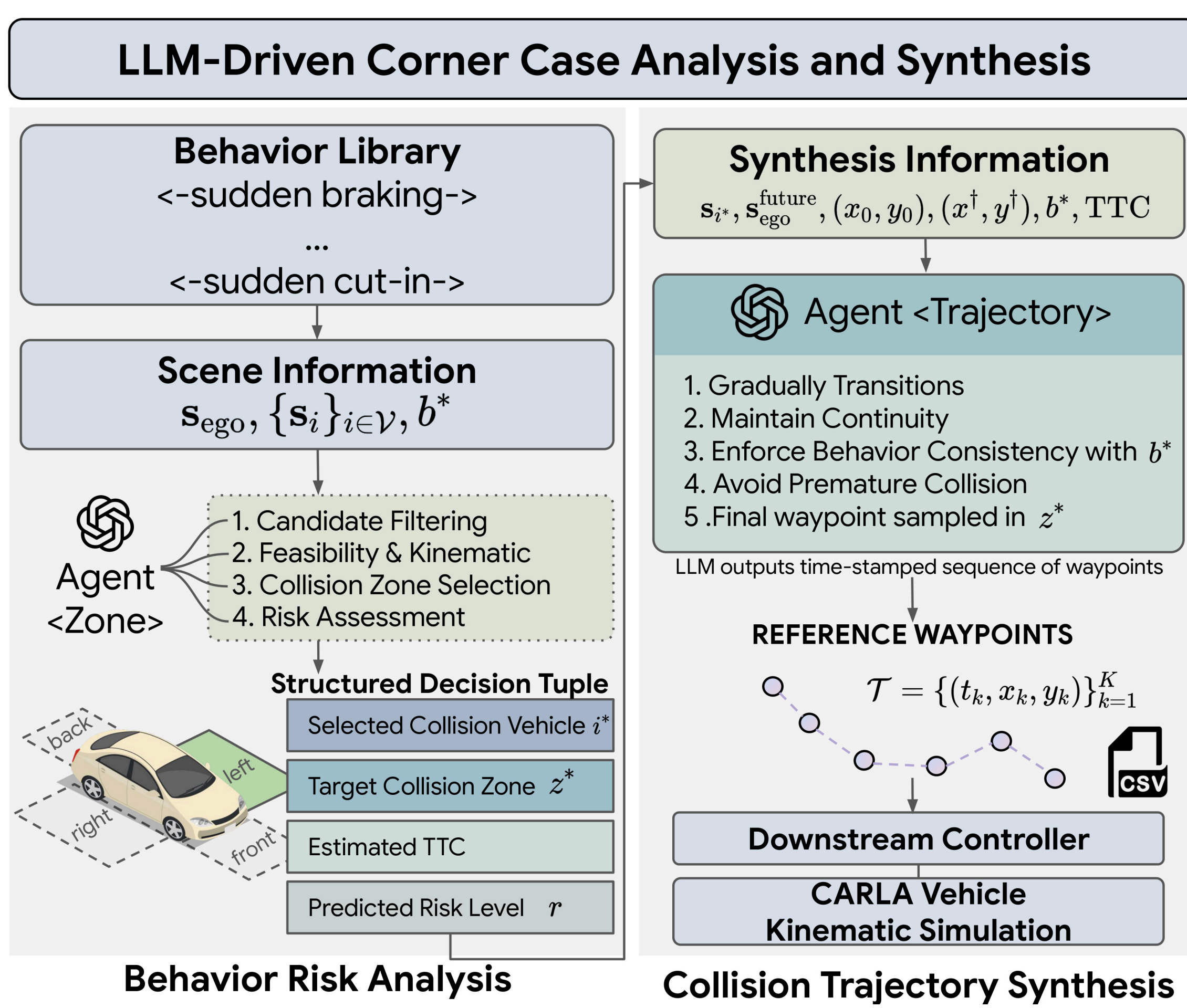
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02 Three modules, one connected pipeline



The complete pipeline



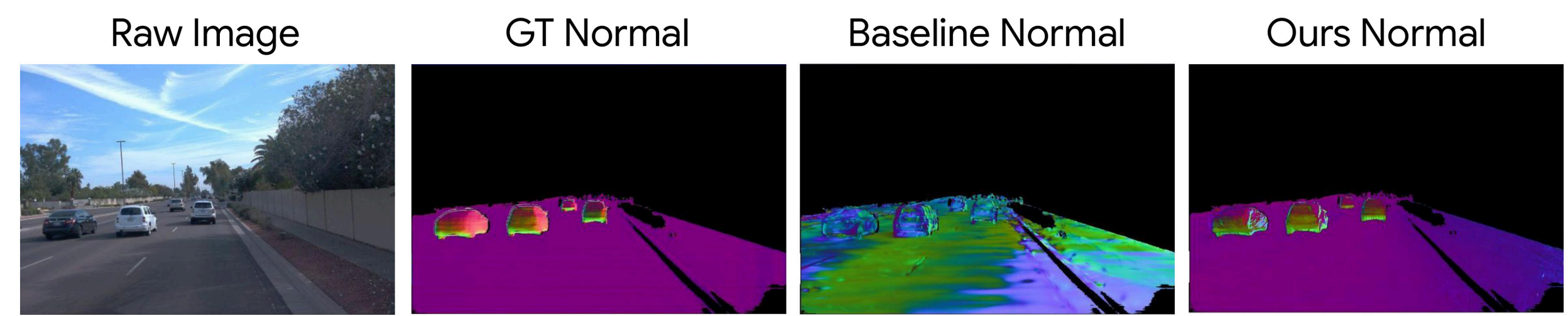
Inside the LLM: choose → plan

01 Motivation & research gap

Prior knowledge- and model-based methods often treat scenes and trajectories in isolation. End-to-end diffusion still struggles with spatiotemporal consistency and physical realism.

Our approach. Connect editable scenes, risk reasoning and physics simulation for realistic corner-case synthesis.

05 Better scene geometry



More consistent surfaces. Geometry constraints reduce normal error from **0.682** to **0.350**, using depth-derived reference normals.

04 From behavior to visual scenarios

Sudden braking (row 1&2) · Cut-in (row 3&4) · Lane drift (row 5&6)

Clearer views of moving cars. Sparse camera views can leave vehicle shapes incomplete. An aligned replacement built from 3D foundation model reduces tearing when a car is viewed from a new angle.

REPRESENT

Rebuild the recorded street as an editable 3D scene. Keep cars separate from the background.

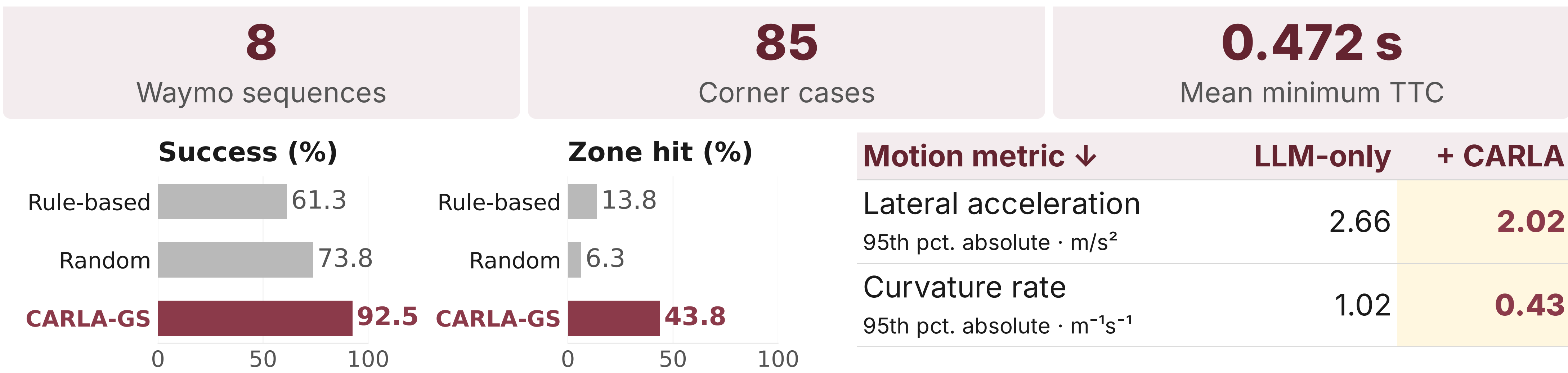
REASON

Use two LLM agents to decide which car should move, where it should go, and which timed path could create a critical interaction.

EXECUTE & RENDER

Follow that path in a CARLA simulator. Send the resulting car positions back into the 3DGS to render the event from the ego vehicle's camera.

03 More critical interactions, smoother motion



What improves? CARLA-GS achieves 92.5% SR and 43.8% zone hit, supporting the multi-agent LLM framework. Executing the planned paths in CARLA lowers these average motion metrics, producing smoother trajectories in the experiments.

TAKEAWAY

Connect realistic scenes, risky behavior planning and simulated motion to create controllable driving corner cases.

LIMITATION

Manual inspection found **29.4% valid LLM outputs**. Planning reliability remains limited.

